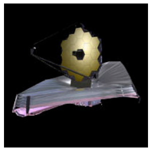

4640 - SKY in 30D: Stellar Kinematic studY in 30 Doradus

Cycle: 3, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Star formation (SF) is a dynamic process that leaves distinctive marks on the kinematics of young stars. These signatures can be used to discriminate between competing theories for cluster assembly. While Gaia's exploration of young star clusters in our Milky Way (MW) provides valuable insights into the star-forming process in well-established spiral galaxies, it cannot access the intense SF rates seen in starbursts, interacting galaxies, and high-redshift regions. We propose an Archival Legacy Program that exploits the unique JWST/ERO dataset of the Large Magellanic Cloud's (LMC) 30Doradus, the brightest starburst in the Local Group. By comparison to existing Hubble Archival data with a 13-year time baseline, we will chart the evolution of this starburst with a resolution comparable to studies of star-forming regions in the MW. This is inaccessible to Gaia due to the distance and severe crowding. But through JWST and HST's high sensitivity, spatial resolution, and focus stability, we will measure projected motions with 2-4 km/s uncertainty for thousands of stars with mass 0.8-17 Msun. The average motions of stellar sub-clusters will be constrained to 1-2 km/s, allowing us to separate distinct stellar components kinematically. Our program will also enable a wealth of other SF investigations by the astronomical community, as well as new studies in unrelated areas such as binary stellar evolution, dust extinction, and LMC kinematics. We will release catalogs of proper motions, photometry, and stellar parameters as High-Level Science Products to the MAST archive. We will post the Jupyter notebooks used for the analysis of the data on GitHub.

OBSERVING DESCRIPTION